

Koushik Sri Sai Varma Datla

(551)-888-0801 • datlakoushik.14@gmail.com • [LinkedIn](#) • [Azure Data Engineer Associate](#)



Professional Summary

Certified Azure Data Engineer with 4+ years of experience in cloud-based **ETL pipeline development**, Data Modelling, Data Integration, Data Migration, Data Extraction, Data Manipulation, Distributed Systems, Data Warehousing, Database Management, Database Design.

Technical Skills

Programming Languages: R, Scala, **SQL**, Spark SQL, T-SQL, Python (**NumPy**, **Pandas**, Matplotlib, **PySpark**), Bash, PowerShell.
Visualization Tools: Tableau, **Power BI**, Excel (Pivot Tables, VLOOKUP, Logical Functions, VBA Macros), Vertica, SAS, SSRS, SSAS.
Big Data: **Snowflake**, **Databricks**, Azure Data Factory, Azure Active Directory, Azure Functions, Apache (Hadoop, Hive, Spark, Kafka, Airflow, NiFi), Redshift, BigQuery, Athena, EC2, Lambda, RDS, Amazon S3, Elastic Search.
Databases & CI/CD: **MySQL**, MongoDB, Microsoft SQL Server, **PostgreSQL**, Postgres, Cosmos DB, **Git**, GitHub, Bitbucket, GitHub Actions, GitLab, Docker, Kubernetes, Terraform, Jenkins, DBT (Data Build Tool).
Software Development: Agile, Scrum, Kanban, Waterfall, Agile Development, CI/CD, continuous Integration, JIRA, BambooHR, Kibana.
API's & Web Services: REST API's, RESTful, Web services, Microservices, Serverless, Presto.
Testing & Quality: Automated Testing, Integration Testing, System Testing, User Acceptance Testing, Debugging.
Data Engineering: ETL, ELT, Data Lake, Sqoop, CSV, XML, JSON, Parquet.

Experience

Data Engineer

January 2023 – Present

Kaiser Permanente, Texas

- Designed **ETL pipeline's** using **Azure Data Factory**, processing over 10 TB of Electronic Health Records (**EHR**) data daily.
- Optimized data processing workflows in **Azure Databricks**, reducing **batch processing** time by 40% and saving \$50,000 annually.
- Developed a **real-time data streaming** using **Azure Event Hub and Databricks**, enabling analytics on patient vitals for 500+ ICU beds.
- Created data quality checks using **PySpark**, improving data accuracy by 25% and reducing downstream analytics errors by 30%.
- Maintained a data lake architecture in **ADLS Gen2**, managing over 50 PB of structured and unstructured healthcare data.
- Designed a **slowly changing dimension (SCD2)** solution for tracking historical changes in provider and patient demographics.
- Implemented a **data lineage** solution, **fault-tolerant** mechanisms providing end-to-end visibility of data flow across the entire data estate, enhancing data governance and facilitating regulatory compliance audits.
- Migrated on-premises **SQL Server databases to Azure SQL Database**, improving SQL query performance by 60%.
- Established data quality checks within the **DBT framework**, increasing data accuracy by 25%; implemented automated alert systems that reduced data-related reporting errors, enhancing confidence in business intelligence outputs.
- Streamlined **CI/CD** using **Azure DevOps**, automating the deployment process, which cuts release cycles from bi-weekly to weekly.

Data Engineer

January 2019 – August 2021

Money Expo, Hyderabad, India

- Developed **ETL processes** using Oracle PL/SQL and SSIS, reducing data processing time by 25% and improving efficiency.
- Migrated data assets to **AWS and Snowflake**, enhancing scalability, and performance for over 100 million records across datasets.
- Automated ETL workflows using Unix/Linux Shell scripting, reducing manual tasks by 80% and improving overall efficiency.
- Introduced **real-time data streaming** with **Apache Kafka**, processing 500,000 data points per second for analytics.
- Proposed a comprehensive **data retrieval optimization** plan, utilizing **MapReduce** and **MySQL** for datasets exceeding 2 TB.
- Leveraged **Alteryx** and **AWS Glue** for ETL processes, increasing transformation speed and data accuracy by 20%.
- Integrated Snowflake with **Azure Data Lake Storage**, enabling secure data sharing and processing 2 billion records monthly.
- Created documentation for **data processes, schemas, and data flows**, improving team onboarding and project efficiency by 30%.
- Reengineered **data extraction** and transformation processes, leading to a measurable increase in **throughput** that empowered data teams to handle 1M+ records daily with improved accuracy and reduced latency.
- Optimized **stored procedures** and queries to enhance data processing speed by 40% and ensure data integrity. Utilized **serialization isolation techniques** to maintain **ACID** properties in complex transactional workflows.

Projects

Cloud-Based ETL Pipeline Development with DBT, Snowflake, and Airflow for Data Transformation and Monitoring.

- Automated 95% of data transformations, reducing ETL processing time by 40%. Integrated AWS SNS and Slack ensured 100% alert accuracy, improving real-time issue tracking for over 100 pipeline runs.

Real-Time IoT Data Processing and Analytics with Databricks, Event Hubs, and Snowflake.

- Initiated the design of a scalable real-time streaming pipeline with Databricks and Event Hubs, processing millions of IoT device data points and reducing data ingestion time by 30%, enabling near-instant analytics in Snowflake.

Education

Master of Science, Robotics | Machine Learning

GPA: 3.70/4.00

Northeastern University, College of Engineering, Boston, MA

Bachelor of Technology

GPA: 4.00/4.00

Manipal Institute of Technology, Karnataka, India